

Defense Portfolio Stability Under Shared Costs and Sampling Uncertainty

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Abstract: An unchanged Pareto frontier can make a simulation look more robust than it is. We examine this problem in a single synthetic hospital model with pathway-specific defenses, six objectives, and 160,000 paired confirmation executions across 400 profile-candidates. Changing the clinical outage definition moves high-capacity event frequency by 14.34 percentage points without changing its observed frontier. An interval calculation retains 72 of 95 efficient portfolios when only that endpoint varies. We then let shared component prices vary and recompute the stochastic objectives through paired, entry-stratified resampling. At $\pm 50\%$ prices, even requiring every upgrade to retain half its nominal incremental price reduces guaranteed retention to 27. Only 12 of the original certificates persist in at least 95% of resamples. A conservative simultaneous bound on population means supports only the free baselines. An external construct audit of 316 public clinical-harm records also identifies services and time periods outside the model. The contribution is a reproducible joint stability study with an exact shared-price retention calculation and explicit separation of deterministic guarantees, resampling frequencies, and statistical confidence. The results identify which assumptions deserve scrutiny before using the simulated tradeoffs to inform a decision.

Index Terms: simulation; defense portfolios; Pareto efficiency; shared costs; sampling uncertainty; reproducibility

1. Introduction

Security investments compete for money, staff time, and operational attention. A simulation can compare many combinations of defenses, but an efficient portfolio is only as persuasive as the assumptions behind its objectives. Reproducing the objective table does not establish that its frontier will survive a different outcome definition, price schedule, or scenario sample.

We investigate these distinctions in a single synthetic hospital ransomware model. The experiment compares 400 profile-candidates on disruption, extreme loss, outage frequency, non-recovery, cost, and burden. An initial check appears reassuring: changing the outage definition leaves two profiles' frontiers unchanged. Yet this check holds both prices and other estimated objectives fixed. Our research question is therefore: *how much of that apparent stability survives when these uncertainties are considered together?*

The answer changes the interpretation of the experiment. The 72 portfolios retained under endpoint uncertainty fall to 27 under a shared-price region that keeps upgrades strictly costly. Scenario resampling provides a further, distinct check; it is not a statistical guarantee. External clinical-harm data reveal where the modeled services and horizon fall short of reported disruption.

The contribution is this joint analysis and its inspectable implementation. Security portfolio simulation-optimization, interval efficiency, and bootstrap selection are established methods [1]–[3]. We combine them in a finite paired experiment, exploit the model's shared component-price structure, and compare the conclusions supported by three kinds of uncertainty analysis. The hospital reanalysis is retrospective. Its plans record which results were already known and when the positive-upgrade-price extension was added.

2. Model and Experimental Design

2.1. Mechanisms and evidence

The model represents one facility as a network of 200–300 digital assets. Incidents enter through a workstation, internet-facing asset, privileged system, medical device, or vendor connection. Compromise, detection, isolation, and restoration evolve in five-minute steps over 72 hours. The resource-constrained, intermediate, and high-capacity profiles are fixed starting environments. Portfolios add controls to that environment; they cannot buy a different capacity profile.

Controls follow the pathway on which they act. Patching mainly reduces exploit-mediated traversal, has little effect on credential-mediated traversal, and has partial effect on vendor pathways. Identity protection has separate coverage and effectiveness on eligible authentication paths. Isolated backups lack a normal traversal path but can suffer an incident-level isolation lapse or residual recovery failure. Restoration either waits for containment or proceeds in parallel in a separate structural experiment. Restored nodes become susceptible again, allowing reinfection.

An evidence register distinguishes observed quantities, contextual evidence, technical assumptions, and normalized decision units. Hospital incident studies establish that disruption occurs [4]; authentication research informs the scope of identity protection [5]. Neither identifies whole-network control effects in this model. Technical effect sizes remain non-identifiable from the available public sources. We sample 10 coefficients over declared ranges once per scenario and share the vector across candidate replays. These ranges express assumptions to examine, not estimates of hospital-sector effectiveness.

2.2. Objectives and paired scenarios

A service is available when at least 60% of its supporting nodes are functional. Weighted service-hours lost

sum unavailable time across EHR, laboratory, pharmacy, imaging, scheduling, identity, and backup/recovery, with weights (1.0, 0.8, 0.9, 0.7, 0.4, 0.9, 0.6). The resulting loss is bounded by 381.6 weighted hours. Only the first four services enter the clinical-outage endpoint. This endpoint measures digital-service availability, not patient outcomes or treatment capacity.

Each candidate has six minimized objectives: mean weighted loss, empirical upper-tail mean loss, sustained-outage frequency, non-recovery by 72 hours, implementation cost, and operational burden. The historical tail statistic averages losses at or above the empirical 90th percentile, including ties. Cost and burden are normalized scenario points. Segmentation, patching, detection, isolation, backups, and identity protection are priced as upgrades above the profile's existing posture. The three profiles have 288, 96, and 16 distinct resolved candidates.

Discovery uses 30 scenarios per profile and 12,000 candidate executions. Confirmation uses 400 fresh scenarios per profile and 160,000 executions. Within a profile, all candidates share event-keyed random inputs. Disjoint scenario-ID blocks separate stages and profiles. Confirmation allocates 80 scenarios to each of five entry categories, defining an equally weighted scenario mixture. The independent units are scenarios; candidate executions are paired observations.

3. Stability Analysis

3.1. Endpoint definitions and interval efficiency

Let N_{pci} count clinical services whose individual continuous outage exceeds two hours for profile p , candidate c , and scenario i . Define $Y_{pci}^{(k)} = \mathbf{1}\{N_{pci} \geq k\}$ for $k = 1, \dots, 4$. The qualifying outages need not overlap. We report all four values of k . The primary endpoint uses $k = 4$. For a fixed population and weighting,

$$\overline{Y^{(1)}} - \overline{Y^{(4)}} = \overline{\mathbf{1}\{1 \leq N \leq 3\}}. \quad (1)$$

The partial-outage fraction is exactly the threshold sensitivity. Using it to justify threshold insensitivity would be circular.

Write $a \prec b$ if a is no worse in every minimized coordinate and strictly better in at least one. The efficient set F contains candidates with no dominator. For independent bounds $L_c \leq f_c \leq U_c$, standard necessary/possible efficiency gives the finite sets

$$G = \{c : \nexists d \neq c, L_d \prec U_c\}, \quad (2)$$

$$P = \{c : \nexists d \neq c, U_d \prec L_c\}. \quad (3)$$

Thus $G = \bigcap F$ and $P = \bigcup F$ over the full Cartesian box [2]. Appendix A gives the short proof. The initial application varies each candidate's outage frequency from its $k = 4$ to $k = 1$ estimate while holding five objectives fixed. This box includes every common- k specification and additional candidate-specific combinations.

3.2. Shared cost and burden uncertainty

Independent candidate cost intervals would discard the model's pricing structure. Instead, write cost as $a_c^\top t$ and burden as $a_c^\top u$, where a_c records purchased upgrades and t, u are shared eight-component price vectors. Upgrade coefficients can be negative when subtracting the price of a baseline ladder rung. Every candidate and profile uses the same component prices within each family.

For radius ρ , prices lie between $(1 - \rho)$ and $(1 + \rho)$ times their nominal values. Basic segmentation cannot exceed least-privilege segmentation, and periodic backups cannot exceed protected backups. We call this the *monotone region*. A second region also requires each of these ladder gaps to retain at least $(1 - \rho)$ times its nominal incremental price. This *positive-gap region* excludes free upgrades. Both are declared stress regions. We evaluate $\rho \in \{0, 0.10, 0.25, 0.50, 0.75\}$.

For each rival d and candidate c , compute

$$m_{dc}^t = \min_{t \in T_\rho} (a_d - a_c)^\top t, \quad m_{dc}^u = \min_{u \in U_\rho} (a_d - a_c)^\top u. \quad (4)$$

Each price region separates into four scalar intervals and two constrained rectangles. Evaluating their vertices gives the exact linear minimum. Combine these minima with the lower differences in the four stochastic objectives. A rival can dominate under some admissible realization exactly when all six minima are nonpositive and at least one is negative. A candidate is guaranteed efficient when no rival satisfies that condition.

This calculation preserves shared prices and costs $O(n^2m)$ for fixed price dimension. For each rival, its outcome, cost, and burden minima can be attained together because those three uncertainty blocks are independent. That suffices for guaranteed retention. It does not produce an exact possibly efficient set under coupled prices: avoiding each rival at a different price need not avoid all rivals at one price.

3.3. Resampling and simultaneous mean bounds

The deterministic calculation fixes estimated outcomes. We next resample whole scenarios within each entry category, preserving candidate pairing. Each of 1000 draws per profile recomputes all four stochastic objectives and both endpoint extremes. We apply every price radius and region to those same draws. Candidate retention frequencies and count quantiles describe this conditional resampling experiment. We do not interpret the discontinuous frontier's bootstrap percentiles as confidence sets for population efficiency.

For a separate finite-sample check, define paired rival-minus-candidate differences in mean loss, non-recovery, and the conservative outage contrast $Y_d^{(4)} - Y_c^{(1)}$. Their ranges have widths $2M, 2, 2$, where $M = 381.6$. Hoeffding's inequality and a union bound [6] give simultaneous one-sided lower bounds

$$\ell_{dc,j} = \overline{D}_{dc,j} - R_j \sqrt{\frac{\log(K/\alpha)}{2n}}, \quad (5)$$

where R_j is the range width, $n = 400$, and $K = 276,048$ includes three means for every ordered distinct pair across all profiles. Independent scenarios with the fixed stratum allocation suffice; comparisons need not be independent. We report $\alpha = 0.10, 0.05, 0.01$, each defining its own simultaneous family.

We certify a candidate only if every rival has a strictly positive lower bound in at least one of these means or in Eq. 4. The rival then cannot dominate, whatever the tail coordinate is. Requiring a strictly worse rival coordinate is essential: ties in all screened coordinates leave room for dominance on the omitted tail objective. This is a sufficient, potentially conservative population-efficiency screen for the declared scenario generator.

3.4. Structural and external checks

The preserved parameter analysis compares rankings between the low and high thirds of each sampled coefficient. A distinct paired structural bank of 150 scenarios per profile compares containment-gated and parallel restoration for 96 selected candidates. These alternatives examine mechanisms that resampling a fixed bank cannot change.

We also audit construct coverage against CIPHER v1.0.1, an independently authored public dataset of reported clinical harms [7]. Before aggregation, we mapped its technical-domain labels to represented, absent, or unspecified model services. We count coded records and distinct reference links, correct one documented time-label typo, and report all categories and leave-one-reference-out influence ranges. Week 2 and First Month annotations extend beyond the model horizon; First Week straddles it. These annotations are not recovery durations. CIPHER records are not independent incidents, and its impact scores are not used as model weights.

Two published engineering problems, RE21 and RE22 [8], provide separate implementation checks for Eqs. 2 and 3. Their pre-recorded design uses 128 finite sampled candidates per problem, three objective-bound radii, an independent frontier oracle, and exact small corner enumerations. The shared-price support function is separately checked against linear programming, including positive-gap regions. Appendix B reports the benchmark settings.

4. Results

4.1. Endpoint agreement

The intermediate and high-capacity profiles have double-digit changes in event frequency but identical frontier membership across all four thresholds. The resource-constrained frontier changes despite its smaller frequency gap (Table 1). The size of an outcome change therefore does not rank the size of a decision-set change.

Across the full endpoint box, 72 of 95 observed efficient candidates remain efficient (Table 2). All 9 high-capacity members are retained, while 7 other candidates could join

TABLE 1. CONFIRMATION EVENT FREQUENCIES (%), THEIR GAP IN PERCENTAGE POINTS, AND FRONTIER JACCARD SIMILARITY BETWEEN $k = 1$ AND $k = 4$.

Profile	$k = 1$	$k = 4$	Gap	Jaccard
Resource-constrained	87.32	83.45	3.87	0.896
Intermediate	57.39	44.79	12.60	1.000
High-capacity	16.00	1.66	14.34	1.000

TABLE 2. ENDPOINT INTERVAL CALCULATION WITH FIVE OBJECTIVES FIXED. G : GUARANTEED EFFICIENT; F_4 : OBSERVED FRONTIER; P : POSSIBLY EFFICIENT.

Profile	Candidates	$ G $	$ F_4 $	$ P $
Resource-constrained	288	42	64	79
Intermediate	96	21	22	41
High-capacity	16	9	9	16

them. Even here, retention is a weaker conclusion than exact frontier invariance.

4.2. Joint price and sampling sensitivity

Adding shared price uncertainty reduces guaranteed retention in every profile (Figure 1). At $\rho = 0.50$, the monotone region retains 21 candidates. Requiring positive upgrade gaps retains 27 (Table 3). The decline is therefore not explained solely by free upgrades at a price-region boundary. The intermediate profile is particularly sensitive to which incremental prices the region permits.

When the stochastic objectives are recomputed from resampled scenarios, 8 original certificates meet the 95% retention-frequency criterion in the monotone region and 12 in the positive-gap region. All radii, per-candidate frequencies, and count distributions are released, including candidates that gain efficiency in some resamples. This result describes which original certificates recur, not a new shortlist of clinically validated defenses.

The simultaneous mean screen is much more conservative. At family $\alpha = 0.05$, its margins are 106.3 weighted hours for loss differences and 27.9 percentage points for binary differences. At $\rho = 0.50$ it certifies 2 portfolios in the monotone region and 3 in the positive-gap region. Those retained are free baselines, protected by their cost and burden coordinates; no purchased portfolio is certified. This remains so at the other two reported alpha levels. The available bank supports useful comparisons, but this distribution-free screen does not resolve their population frontier membership.

The earlier 500 random-price draws also change frontier membership in every profile. The new calculation answers the stronger universal question over the specified price region rather than relying on sampled price draws.

4.3. Structural and clinical gaps

The structural experiment finds 21 candidates efficient only with containment-gated restoration and 3 only with

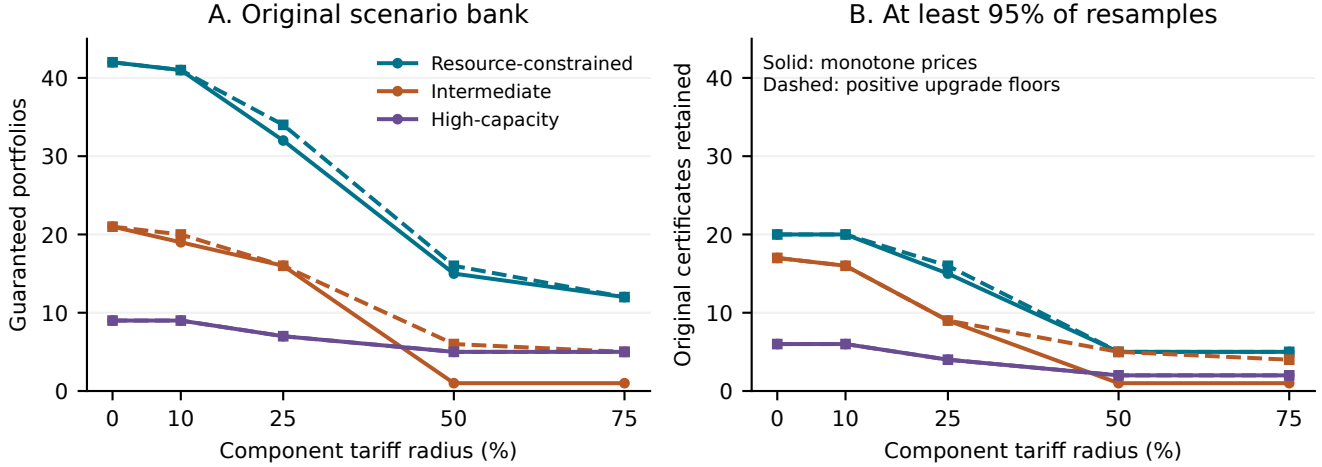


Figure 1. Joint endpoint and price sensitivity across all planned radii. Panel A holds the original outcome estimates fixed. Panel B counts members of the original endpoint-guaranteed set retained in at least 95% of 1000 paired stratified resamples. The percentage in panel B is a resampling frequency, not a confidence level. Positive-gap prices preserve at least $(1 - \rho)$ of each nominal ladder increment.

TABLE 3. RETENTION AT $\pm 50\%$ SHARED PRICES. G USES THE ORIGINAL SCENARIO BANK; B_{95} COUNTS ORIGINAL ENDPOINT CERTIFICATES RETAINED IN AT LEAST 95% OF RESAMPLES.

Profile	Monotone		Positive gaps	
	G	B_{95}	G	B_{95}
Resource-constrained	15	5	16	5
Intermediate	1	1	6	5
High-capacity	5	2	5	2

TABLE 4. CIPHER CONSTRUCT COVERAGE. COUNTS DESCRIBE CODED RECORDS, NOT INDEPENDENT INCIDENTS. A REFERENCE CAN CONTRIBUTE TO SEVERAL ROWS.

Coverage class	Records	References	%
Represented service	229	32	72.5
Absent explicit service	85	19	26.9
Unspecified	2	1	0.6

parallel restoration, within the selected candidate bank. Intermediate mean loss rises from 46.93 to 107.41 weighted hours when restoration can begin before containment. Earlier restoration can expose susceptible nodes to a still-active compromise process. High-capacity frontier membership stays unchanged, although objective values move. Parameter ranking correlations, whose worst evaluated value is 0.606, do not settle this structural question.

CIPHER contains 316 records linked to 36 distinct references. Of these, 85 (26.9%) concern technical domains without an explicit modeled service, including communications, infrastructure, operating rooms, and telemetry (Table 4). Another dimension is time: 97 records (30.7%) carry Week 2 or First Month annotations. Removing one reference at a time gives shares of 25.7–30.3% for absent services and 17.4–31.7% for these later annotations. The latter range

shows substantial source concentration. Even a represented service label does not establish that its associated patient harm is modeled.

The existing external comparison register also exposes a timing mismatch. The restoration model is slower than the short technical interruption in the CrowdStrike comparison [9] and cannot reproduce the extended clinical disruption in ransomware recovery records [10]. These targets measure different processes, so they cannot be pooled into one fitted restoration rate. Of nine registered comparisons, one is a numeric failure, two are qualitative failures, one shows loose numeric consistency, three are not scored, and two are structurally unaddressable. None establishes calibrated hospital recovery dynamics.

4.4. Interpretation checks

Paired analysis and candidate-set checks resolve two additional errors. The high-capacity candidate with lowest observed mean loss saves 1.95 weighted service-hours against its baseline. Its paired stratified SE is 0.333, versus 0.944 with covariance omitted. The approximate simultaneous interval across all 397 baseline contrasts is [0.66, 3.25] hours. This supports a within-model mean difference without certifying the candidate’s six-objective population efficiency.

The full-space frontier has 95 candidates, compared with 57 within the frozen discovery shortlist, which omits 38 efficient candidates. Every resample contains an omission; the rounded central range is 29–49. The original bank has no candidate efficient only because of shortlist restriction, but at least one such artifact appears in 42.7% of resamples.

Finally, the preserved numeric audit passes on three historical interpretation errors. Explicit assertions reject all 3 cases, accept 8 correct controls, and reject or flag invalid all 8 recorded mutations. Table 5 provides the worked exam-

ples; Appendix C describes the checks. This is a regression evaluation of specified assertions, not a measure of unseen-error detection.

5. Relation to Prior Work

Kiesling et al. combine security-control models, multi-objective simulation-optimization, and portfolio selection [1]. Hladík studies efficiency under interval uncertainty [2]. Our finite-box formulas are direct specializations of that established idea. Their role is to make a limited stability claim explicit and computable; the shared-price calculation preserves the dependency structure of the portfolio cost model.

Simulation ranking and selection also has a substantial literature. Currie and Monks’s BootComp uses bootstrap comparisons, permits common random numbers, and includes a hospital ward application [3]. Andradóttir and Lee develop Pareto-set selection procedures with guaranteed correct-selection probabilities under independent or correlated sampling [11]. Our fixed-bank study does not propose a replacement selection algorithm. It compares deterministic retention, conditional resampling frequency, and a conservative simultaneous mean screen on the same candidate set.

The design follows the transparency goals of STRESS reporting [12], common-random-number practice [13], and explicit separation of prior plans from retrospective analysis [14]. Multiverse and specification-curve studies motivate examining analytical choices [15], [16]; our study does not implement their joint inference procedures. Statistical consistency checks [17], metamorphic simulation tests [18], and the ABE-Ralph experimental-fidelity preprint [19] provide context for the interpretation assertions. The scientific addition is the documented change in conclusions when several assumptions are examined together, supported by an auditable implementation and public construct evidence.

6. Discussion

The main implication is about where to spend validation effort. Endpoint agreement was the most reassuring result, but shared prices removed most of its guarantees. Maintaining positive upgrade prices softened this decline without eliminating it. A decision-maker using this model would need defensible incremental cost and burden ranges before treating frontier membership as a stable reason to prefer a portfolio.

The model’s temporal dynamics require separate work. Short technical restoration, prolonged clinical disruption, and a time-annotated harm record are different outcomes. The external sources document relevant consequences but do not identify the transitions or control coefficients that generate them here. CIPHER adds an independently authored check of construct coverage; it is not an independent defense experiment. The single-facility graph also excludes correlated multi-hospital incidents and regional spillover.

Future applied validation would need local service dependencies, containment and restoration timelines, and measured implementation costs.

The mathematical scope is narrower and precise. End-point intervals permit candidate-specific combinations that a common threshold cannot generate. Price regions are chosen stress assumptions. The bootstrap conditions on the observed scenario bank, and the selected-candidate structural experiment does not search every possible mechanism. The simultaneous mean bound requires independent scenario draws and is deliberately conservative; failure to certify is not evidence of inefficiency. Its retained free baselines illustrate why Pareto efficiency alone is not a recommendation.

The work uses synthetic trials and public sources, with no recruited participants or private patient records. The CIPHER archive is preserved under its data license. The earlier THREAT analysis uses a normalized repository preview export, not the original deposited file, and is retained as descriptive scope evidence. Published engineering functions test implementation on independently defined problems, while evaluation remains within this research workflow. Scientific transport and independent replication therefore remain distinct from the verified artifact.

7. Conclusion

The apparent stability of a simulated defense frontier depends on what a sensitivity check allows to change. End-point uncertainty retains 72 observed efficient portfolios in this experiment. Shared prices reduce that count to 27 even when upgrades remain strictly costly, and only 12 original certificates persist in at least 95% of paired resamples. A conservative population check supports only free baselines. Public clinical-harm records identify additional service and temporal gaps. Together, these findings turn a reassuring end-point result into a more useful account of which assumptions constrain the decision.

Artifact Availability

The artifact contains the original trial banks and frozen protocols, retrospective plans and amendments, licensed external inputs, analysis code, generated result tables and figures, and verification commands. The confirmation bank is stored compressed and restored losslessly. The identified repository link is omitted from this review copy.

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TABLE 5. WORKED INTERPRETATION AUDIT. HISTORICAL SENTENCES ARE SHORTENED FAITHFULLY FROM THE ARCHIVED MANUSCRIPT. ASSERTIONS ENCODE THE STATED SCOPE, UNIT, AND BOUND; A PASSING NUMERIC-MACRO CHECK ALONE DID NOT TEST THESE RELATIONS.

Historical statement	Numerical or scope evidence	Executable assertion	Corrected conclusion
Endpoint values move by roughly three percentage points across k .	High-capacity confirmation gap: 14.34 percentage points.	All confirmation profiles present; $\max_p \Delta_p \leq 4$ pp. Fails.	Frequency changes are profile-specific; unchanged frontiers do not imply small frequency gaps.
The high-capacity panel spans under one service-hour.	Baseline-to-best mean difference: 1.95 weighted service-hours.	High-capacity mean-loss range < 1 hour. Fails.	The observed baseline-to-best span exceeds one hour; paired uncertainty is reported separately.
The figure presents discovery results.	Plotted values come from the confirmation objective table.	Source stage equals discovery. Fails.	Identify confirmation as the plotted stage and report discovery separately.

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LLM Usage Statement

AI tools assisted with literature review, code development and revision, verification, and manuscript preparation.

Appendix A. Efficiency Proofs and Numerical Scope

For Eqs. 2 and 3, if $L_d \prec U_c$, placing d at its lower bound and c at its upper bound makes c dominated. Conversely, a realized dominator implies $L_d \prec U_c$. If $U_d \prec L_c$, d dominates c throughout the box. Otherwise, place c at L_c and every rival at its upper bound; no rival dominates c . Independent box coordinates permit these assignments. Strict dominance retains exact ties.

For Eq. 4, minimizing a linear function on a constrained rectangle requires only its vertices: admissible box corners and the intersections with the minimum-gap boundary. Uncoupled coordinates choose the lower or upper endpoint according to the coefficient’s sign. Cost, burden, and outcome minima can be realized simultaneously for a fixed rival. The existence of any such rival disproves universal retention; its absence proves retention. Price calculations use integer upgrade coefficients and round cancellation residues to 12 decimal places. Independent linear-programming checks agree to 10^{-10} .

TABLE 6. ALL PRICE RADII. ENTRIES GIVE COUNTS IN R/I/H PROFILE ORDER. B_{95} COUNTS ORIGINAL CERTIFICATES MEETING THE RESAMPLING-FREQUENCY CRITERION. THE POSITIVE-GAP EXTENSION WAS RECORDED AFTER THE FIRST RUN.

Radius %	Monotone		Positive gaps	
	G	B_{95}	G	B_{95}
0	42/21/9	20/17/6	42/21/9	20/17/6
10	41/19/9	20/16/6	41/20/9	20/16/6
25	32/16/7	15/9/4	34/16/7	16/9/4
50	15/1/5	5/1/2	16/6/5	5/5/2
75	12/1/5	5/1/2	12/5/5	5/4/2

TABLE 7. ALL PLANNED ENGINEERING SETTINGS, WITH 128 FINITE SAMPLED CANDIDATES PER PROBLEM. RADIUS SCALES THE SAMPLED OBJECTIVE RANGE.

Problem	Radius	$ G $	$ F_0 $	$ P $
RE21	1%	9	15	19
RE21	5%	0	15	65
RE21	10%	0	15	120
RE22	1%	0	12	91
RE22	5%	0	12	120
RE22	10%	0	12	125

The population screen follows from Hoeffding’s inequality for independent bounded observations, including non-identically distributed observations in fixed strata. A union bound over K comparisons gives simultaneous coverage of at least $1 - \alpha$. If a lower bound is strictly positive for every rival in some coordinate, none can dominate the candidate. No tail-mean concentration claim is required. An equal screened vector cannot be discarded, since an omitted tail coordinate could break the tie.

Appendix B. Engineering and Decision Diagnostics

RE21 models a four-bar truss; RE22 models a reinforced-concrete beam and retains its constraint-violation objective [8]. Independent intervals perturb all objectives by $\pm 1\%$, $\pm 5\%$, $\pm 10\%$ of their finite sampled ranges, with non-negative lower bounds. All 6,000 perturbation frontiers contain G and lie within P . For the first eight candidates per problem, 512 total corner matrices with only the second objective uncertain reproduce the exact frontier intersection and union. 5 of six settings have empty guaranteed sets. These checks concern finite tables and chosen bounds, not the continuous problems’ global frontiers.

The budget example selects the lowest observed mean loss within each profile’s point budget. The artifact identifies the portfolios and their burdens. Another objective or a burden constraint can favor another choice; zero observed events do not imply zero risk.

Tied minima explain the large intermediate single-objective maximum. Adding an objective can break ties and remove a previously efficient candidate, so cardinality need not increase with dimension.

TABLE 8. ILLUSTRATIVE MEAN-LOSS SELECTION WITHIN EACH DECLARED POINT BUDGET. FREE AND SELECTED COLUMNS GIVE MEAN WEIGHTED SERVICE-HOURS LOST; THE LAST COLUMN GIVES THE SELECTED CANDIDATE’S $k = 4$ FREQUENCY.

Profile	Budget	Free	Selected	Event %
Resource-constrained	5	322.13	156.99	94.50
Intermediate	10	179.34	24.34	21.50
High-capacity	15	15.02	13.07	0.00

TABLE 9. FRONTIER SIZES OVER EVERY OBJECTIVE SUBSET OF A GIVEN DIMENSION, SHOWN AS MINIMUM/MEDIAN/MAXIMUM. TIED OPTIMA ARE RETAINED.

Dimensions	R	I	H
1	1/1/3	1/1/47	1/1.5/8
2	1/6/16	1/3/18	1/2/9
3	1/18.5/45	1/14/22	1/4.5/9
4	6/45/53	1/21/22	1/6/9
5	28/54/64	14/22/22	5/9/9
6	64/64/64	22/22/22	9/9/9

Appendix C. Interpretation and Reproduction

Each registered assertion specifies a source, population, expected groups, statistic, unit, quantifier, relation, and bound. Missing groups or unsupported operations stop verification. The eight correct assertions, three historical contradictions, and eight single-field mutations were constructed after reviewing the manuscript. They test specified behavior rather than estimate general detection accuracy.

For paired mean-loss comparisons, let D_{hi} be the candidate difference in entry stratum h , with weight $w_h = n_h/n$. Then

$$\widehat{\text{Var}}(\widehat{\Delta}) = \sum_h w_h^2 s_{D,h}^2 / n_h, \quad s_{D,h}^2 = s_{a,h}^2 + s_{b,h}^2 - 2s_{ab,h}. \quad (6)$$

Setting $v_h = w_h^2 s_{D,h}^2 / n_h$ gives approximate degrees of freedom $(\sum_h v_h)^2 / \sum_h [v_h^2 / (n_h - 1)]$ for the family-adjusted paired t intervals.

Original raw banks and protocol digests remain unchanged. Added analyses start from committed clean source states, with code, environment, and file hashes recorded in manifests. The reproduction guide documents commands and historical provenance, including an earlier dirty source state. Automated checks cover specified relationships; an unregistered interpretation can still escape them.